

Def.

Let V be a real or complex vector space.

A function $f: V \times V \rightarrow F$ is called a sesqui-linear form if it satisfies that for all $\alpha, \beta, \gamma \in V$ and $c \in F$,

$$\begin{cases} f(c\alpha + \beta, \gamma) = c \cdot f(\alpha, \gamma) + f(\beta, \gamma) \\ f(\alpha, c\beta + \gamma) = \overline{c} \cdot f(\alpha, \beta) + f(\alpha, \gamma) \end{cases}$$

In this page, we call the f as form, for

Thm.

Let V be a finite dimensional inner product space and $f: V \times V \rightarrow F$ be a form. Then, there is a unique linear operator T on V such that

$$f(\alpha, \beta) = \langle T\alpha, \beta \rangle \text{ for all } \alpha, \beta \in V.$$

Moreover, the map $f \mapsto T$ is isomorphism from space of forms onto $\mathcal{L}(V, V)$.

Proof. Let $\beta \in V$ be fixed. Then, a map $\alpha \mapsto f(\alpha, \beta)$ is a linear map on V .

By the Riesz-Representation theorem, for an orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$, there exists a unique vector $\beta' = \overline{f(\alpha_1, \beta)} \cdot \alpha_1 + \dots + \overline{f(\alpha_n, \beta)} \cdot \alpha_n$ such that

$$\langle \alpha, \beta' \rangle = f(\alpha, \beta) \text{ for all } \alpha \in V.$$

Define $U: V \rightarrow V; \beta \mapsto \beta'$. Then, given $\alpha, \beta, \gamma \in V$ and $c \in F$,

$$\langle \alpha, U(c\beta + \gamma) \rangle = f(\alpha, c\beta + \gamma) = \overline{c} \cdot f(\alpha, \beta) + f(\alpha, \gamma) = \overline{c} \cdot \langle \alpha, U(\beta) \rangle + \langle \alpha, U(\gamma) \rangle = \langle \alpha, cU(\beta) + U(\gamma) \rangle$$

Since the α was arbitrary, it proves U is linear.

Moreover, since V is finite-dimensional, so the adjoint exists, put $T = U^*$.

Consequently, $f(\alpha, \beta) = \langle \alpha, U(\beta) \rangle = \langle T\alpha, \beta \rangle$. $\square$ Existence.

Corollary. Given forms $f, g: V \times V \rightarrow F$, define

$$\langle f, g \rangle \stackrel{\text{def}}{=} \text{Tr}(T_f T_g^*) \stackrel{(*)}{=} \sum_{j,k} f(\alpha_k, \alpha_j) \cdot \overline{g(\alpha_k, \alpha_j)}$$

is an inner product on $V \times V$ such that $(*)$, for any orthonormal basis $\{\alpha_1, \dots, \alpha_n\}$ for V .

Proof. Given orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$, let $A = [T_f]_{\mathcal{B}}$ and $B = [T_g]_{\mathcal{B}}$.

Then, by the orthonormality and the above Thm,

$$A_{jk} = \langle T_f(\alpha_k), \alpha_j \rangle = f(\alpha_k, \alpha_j) \text{ and } B_{jk} = \langle T_g(\alpha_k), \alpha_j \rangle = g(\alpha_k, \alpha_j).$$

Moreover, $[T_f \circ T_g^*]_{\mathcal{B}} = [T_f]_{\mathcal{B}} \cdot [T_g^*]_{\mathcal{B}} = AB^*$, so $\text{Tr}(AB^*) = \sum_{j,k} A_{jk} \cdot \overline{B_{jk}}$ gives $(*)$.

Def. Let V be a fin. dim inner product space, with basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$.

For a given form $f: V \times V \rightarrow F$, the matrix with entries

$$A_{jk} = f(\alpha_k, \alpha_j)$$

is called the matrix of f in the ordered basis $\mathcal{B}$.

Observe that: for a form $f: V \times V \rightarrow F$ and orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$.

Then, $f\left(\sum_s x_s \alpha_s, \sum_r y_r \alpha_r\right) = \sum_{r,s} \overline{y_r} A_{rs} x_s$, or

$$f(\alpha, \beta) = Y^* A X \text{ where } X = [\alpha]_{\mathcal{B}}, Y = [\beta]_{\mathcal{B}}.$$

Thm. Let V be a fin. dim Complex inn. Prod. space.

For any form on $V \times V$, there is an orthonormal basis for V , in which the matrix of f is upper-triangular.

Proof. Since V is fin. dim. Complex. inn. Prod. space, for linear operator $T: V \rightarrow V$ s.t.

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle,$$

it has an orthonormal basis $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$ for V s.t. $[T]_{\mathcal{B}}$ is upper-triangular. So,

$$f(\alpha_k, \alpha_j) = \langle T(\alpha_k), \alpha_j \rangle = 0 \text{ if } j > k.$$

Def. A form f on V is called Hermitian if

$$f(\alpha, \beta) = \overline{f(\beta, \alpha)} \quad \text{for all } \alpha, \beta \in V.$$

Note: if there is a linear operator T on V such that $f(\alpha, \beta) = \langle T(\alpha), \beta \rangle$, then $\overline{f(\beta, \alpha)} = \langle \alpha, T(\beta) \rangle = \langle T^*(\alpha), \beta \rangle$, so,

f is Hermitian if and only if T is Hermitian.

If T is Hermitian, then $\langle T(\alpha), \alpha \rangle$ is real for all $\alpha \in V$, so $f(\alpha, \alpha)$ is real.

Lemma. Let V be a Complex inner product space, and f is a form on V such that $f(\alpha, \alpha)$ is real, for all $\alpha \in V$.

Then, f is Hermitian.

Proof. Given $\alpha, \beta \in V$,

$$f(\alpha + \beta, \alpha + \beta) = f(\alpha, \alpha) + f(\beta, \beta) + f(\alpha, \beta) + f(\beta, \alpha)$$

$$f(\alpha + i\beta, \alpha + i\beta) = f(\alpha, \alpha) + f(i\beta, i\beta) - i \cdot f(\alpha, \beta) + i f(\beta, \alpha)$$

By the assumption, $f(\alpha, \beta) + f(\beta, \alpha)$ and $-i \cdot f(\alpha, \beta) + i \cdot f(\beta, \alpha)$ are real. Thus,

$$f(\alpha, \beta) + f(\beta, \alpha) = \overline{f(\alpha, \beta)} + \overline{f(\beta, \alpha)} \quad \text{and} \quad -i \cdot f(\alpha, \beta) + i \cdot f(\beta, \alpha) = i \cdot \overline{f(\alpha, \beta)} - i \cdot \overline{f(\beta, \alpha)}$$

Multiplying i to the right, then we obtain

$$f(\alpha, \beta) - f(\beta, \alpha) = -\overline{f(\alpha, \beta)} + \overline{f(\beta, \alpha)}$$

so, $2f(\alpha, \beta) = 2 \cdot \overline{f(\alpha, \beta)}$, or f is Hermitian. $\square$

Combining the above note with this Lemma if V is fin. dim, then

T is Hermitian if and only if $\langle T(\alpha), \alpha \rangle$ is real.

Principal Axis Theorem.

Thm. For every Hermitian form f on a fin. dim inn. prod space V , there exists an orthonormal basis $\beta = \{\alpha_1, \dots, \alpha_n\}$ such that

f is represented by a diagonal matrix with real entries.

Proof. Since V is fin. dim, there exists a linear operator T on V s.t.

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle.$$

And, by the lemma above, f is Hermitian iff T is Hermitian, so there is an orthonormal basis for V which is consisted by eigenvector of T . Denote $\beta = \{\alpha_1, \dots, \alpha_n\}$, and C_i be the eigenvalue corresponding to α_i . Then,

$$f(\alpha_k, \alpha_i) = \langle T(\alpha_k), \alpha_i \rangle = C_k \cdot \delta_{ki},$$

this proved. (C_k are real by T is Hermitian). ~~Q~~

Note: Under the condition hold,

$$f\left(\sum x_i \alpha_i, \sum y_k \alpha_k\right) = \sum_i C_i x_i \overline{y_i}$$

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Def.

A sesqui-linear form f is said to be non-degenerate if: for fixed $\alpha \in V$,
if $f(\alpha, \beta) = 0$ for all $\beta \in V$, then $\alpha = 0$.

Prop.

Let $f: V \times V \rightarrow F$ be a form on an inner product space V .

f is non-degenerate if and only if the associated linear T_f is non-singular.

Proof. Suppose that f is non-degenerate. To show T_f is non-singular, ($\ker T_f = \{0\}$)
assume that $T_f(\alpha) = 0$. That is, for any $\beta \in V$,

$$f(\alpha, \beta) = \langle T_f(\alpha), \beta \rangle = \langle 0, \beta \rangle = 0$$

So, $\alpha = 0$ by definition. Conversely, if T_f is non-singular, then for any non-zero $\alpha \in V$,

$$f(\alpha, \beta) = \langle T_f(\alpha), \beta \rangle \neq 0 \text{ if } \beta = T_f(\alpha)$$

which proves f is non-degenerate, $\square$

Prop.

Let $f: V \times V \rightarrow F$ be a non-degenerate form on a finite dimensional ^{inn. prod} space V .
Given linear functional L , there exists one and only one vector $\beta \in V$ such that
$$L(\alpha) = f(\alpha, \beta) \text{ for all } \alpha \in V.$$

Proof By the theorem above, f has invertible associated linear map T such that
$$f(\alpha, \gamma) = \langle T(\alpha), \gamma \rangle \text{ for all } \alpha, \gamma \in V.$$

Meanwhile, by the Riesz Representation theorem, for the given L ,
there exists a unique vector $\beta' = L(\alpha_1) \cdot \alpha_1 + \dots + L(\alpha_n) \alpha_n$, ($\{\alpha_1, \dots, \alpha_n\}$ is orthonormal)
$$L(\alpha) = \langle \alpha, \beta' \rangle \text{ for all } \alpha \in V.$$

Now, let $\beta = (T^*)^{-1}(\beta')$, then

$$f(\alpha, \beta) = \langle T(\alpha), \beta \rangle = \langle T(\alpha), (T^*)^{-1}(\beta') \rangle = \langle \alpha, (T^*) \circ (T^*)^{-1}(\beta') \rangle = \langle \alpha, \beta' \rangle = L(\alpha).$$

And, the uniqueness is given by: $f(\alpha, \beta_1) = L(\alpha) = f(\alpha, \beta_2) \Rightarrow f(\alpha, \beta_1 - \beta_2) = 0$ and
Since f is non-degenerate, so $\beta_1 - \beta_2$ must be zero. (Let $\alpha = \beta_1 - \beta_2$) ~~QED~~

Prop.

Let $f: V \times V \rightarrow F$ be a non-degenerate form on a finite dimensional inner product space V .
Given a linear operator $S: V \rightarrow V$, there exists an adjoint S' relative to f .
(That is, $f(S\alpha, \beta) = f(\alpha, S'\beta)$ for all $\alpha, \beta \in V$).

Proof